Diffusion in Biology

Exam code: 0610 & 0970

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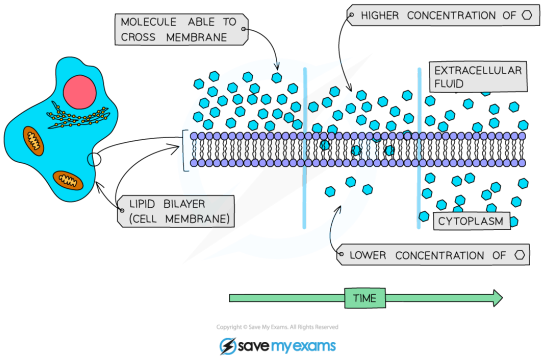
Diffusion

What is Diffusion in Biology?

- Diffusion is the **movement of molecules** from a region of its **higher** concentration to a region of its **lower** concentration
- Molecules **move down a concentration gradient**, as a result of their **random movement**



Diffusion across the cell membrane

- For **living cells**, the principle of the movement down a concentration gradient is the same, but the cell is

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

this means it allows some molecules to cross easily, but others with difficulty or not at all
- The simplest sort of selection is based on the size of the molecules
- Diffusion helps living organisms to:
  - obtain many of their **requirements**
  - get rid of many of their **waste products**
  - carry out **gas exchange for respiration**

Examples of diffusion in living organisms

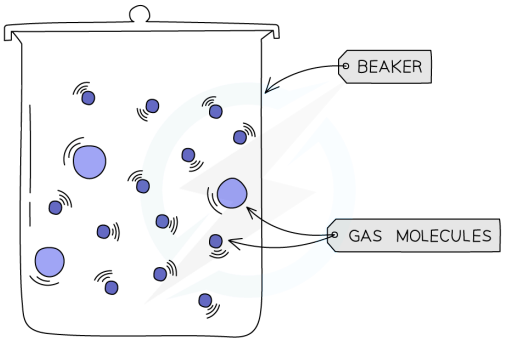
- You will need to learn examples of substances that organisms obtain by diffusion
- Don't forget that **plants require oxygen for respiration at all times**, as well as **carbon dioxide for photosynthesis** when conditions for photosynthesis are right (e.g. enough light and a suitable temperature)

Examples of Diffusion Table

SITE	MOLECULES MOVING	FROM	TO
SMALL INTESTINE	DIGESTED FOOD PRODUCTS – GLUCOSE, AMINO ACIDS, FATTY ACIDS AND GLYCEROL ETC.	LUMEN OF SMALL INTESTINE	BLOOD / LYMPH IN VILLI FOUND COVERING SMALL INTESTINE WALLS
LEAF	OXYGEN	AIR SPACES BETWEEN MESOPHYLL CELLS	MITOCHONDRIA IN ALL CELLS
LEAF	CARBON DIOXIDE	AIR SPACES BETWEEN MESOPHYLL CELLS	CHLOROPLASTS IN MESOPHYLL CELLS
LEAF	WATER VAPOUR	STOMATAL PORES	AIR OUTSIDE STOMATA
LUNGS	OXYGEN	ALVEOLAR AIR SPACE	BLOOD IN CAPILLARIES AROUND ALVEOLI
LUNGS	CARBON DIOXIDE	BLOOD IN CAPILLARIES AROUND ALVEOLI	ALVEOLAR AIR SPACE

Where does the energy for diffusion come from?

- All particles move randomly at all times
- This is known as Brownian motion
- The energy for diffusion comes from the kinetic energy of this random movement of molecules and ions



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